WHAT2WATCH

MOVIE RECOMMENDATION SYSTEM

PROJECT REPORT

This project is an AI/ML-based Movie Recommendation System that uses the TMDB API to suggest movies based on user input. It demonstrates real-time recommendation functionality and practical application of intelligent systems.

Name : Ananya Gupta
Reg No. : 25BCE10923
Branch : BTech CSE Core
Year : 1st Year
Course : Fundamentals in AI and ML
Course Code : CSA2001

INTRODUCTION

With the rapid growth of online streaming platforms, picking the right movie can often feel overwhelming. Instead of enjoying a film, users often end up spending too much time deciding what to watch, which is where recommendation systems come in—they help by suggesting movies based on a user's interests.

In this project, a Movie Recommendation System is created using the TMDB (The Movie Database) API. Users simply type in the name of a movie, and the system fetches a list of similar movie recommendations directly from the TMDB database using API calls.

The application is built as a Command Line Interface (CLI) using Python, making it simple, lightweight, and easy to use, while also giving a clear understanding of how real-world recommendation systems actually work.

MOTIVATION OF THE PROJECT

The motivation behind this project includes:

- Understanding real-world API integration.
- Learning how recommendation systems work in streaming platforms.
- Implementing Python networking using requests library.
- Gaining practical exposure to external databases like TMDB.
- Building an interactive AI-based application.

This project bridges theoretical AI concepts with real-world implementation using live movie data.

PROBLEM STATEMENT

The rapid expansion of online movie databases and streaming platforms has led to an exponential increase in available content. This creates challenges for users in identifying relevant movies that match their interests. The key problems addressed in this project are:

- Information overload due to large-scale movie databases.
- Lack of an automated mechanism to retrieve related movies efficiently.
- Inefficiency of manual search methods in discovering similar content.
- Need for intelligent systems that provide personalized recommendations.
- Requirement for real-time data retrieval and processing using web-based services.

To address these challenges, the project aims to design and implement an AI/ML-based Movie Recommendation System that interacts with an external API, processes JSON responses, and generates relevant movie suggestions dynamically.

OBJECTIVE OF THE PROJECT

The objectives of this project are:

- To develop a movie recommendation system using TMDB API.
- To implement API integration using Python.
- To handle JSON responses from web services.
- To design a user-friendly CLI-based interface.
- To fetch and display top 5 recommended movies.

EXISTING METHODS

1. Content-Based Filtering
Uses item features to recommend similar content.

2. Collaborative Filtering
Uses user behavior and ratings.

3. API-Based Recommendation (Used in This Project)
Uses third-party databases that already implement advanced recommendation algorithms.
This project uses TMDB's built-in recommendation system via API.

PROS AND CONS OF THE STATED METHODS

API-Based Recommendation (Used)

Pros:

- Uses real-time updated data.
- No need to build ML model manually.
- High-quality recommendations.
- Access to large movie database.

Cons:

- Depends on internet connection.
- Dependent on external API availability.
- API rate limits may apply.

Content-Based Filtering

Pros:

- Fully independent system.
- Customizable algorithm.

Cons:

- Requires dataset preprocessing.
- Limited to available data.

HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements

- Minimum 4GB RAM
- Standard PC/Laptop
- Internet connection

Software Requirements

- Python 3.x
- VS Code / IDLE / Any Python IDE
- Required Library:
  - requests

METHODOLOGY AND GOAL

Methodology

1. User inputs movie name.
2. System sends request to TMDB search API.
3. Extract movie ID from JSON response.
4. Send another request to TMDB recommendation endpoint.
5. Retrieve top 5 recommended movies.
6. Display results in CLI interface.

Goal

The goal of this project is to develop an intelligent movie recommendation system that integrates with a real-world API and provides accurate, real-time movie suggestions in an efficient and user-friendly manner.

FUNCTIONAL MODULES DESIGN AND ANALYSIS

1. Movie Search Module (search_movie)
   - Sends request to TMDB search endpoint.
   - Retrieves movie ID.
   - Handles errors if movie not found.

2. Recommendation Module (get_recommendations)
   - Sends request using movie ID.
   - Extracts recommended movies.
   - Returns top 5 results.

3. User Interface Module (main)
   - Displays menu options.
   - Takes user input.
   - Calls respective functions.
   - Handles invalid choices.

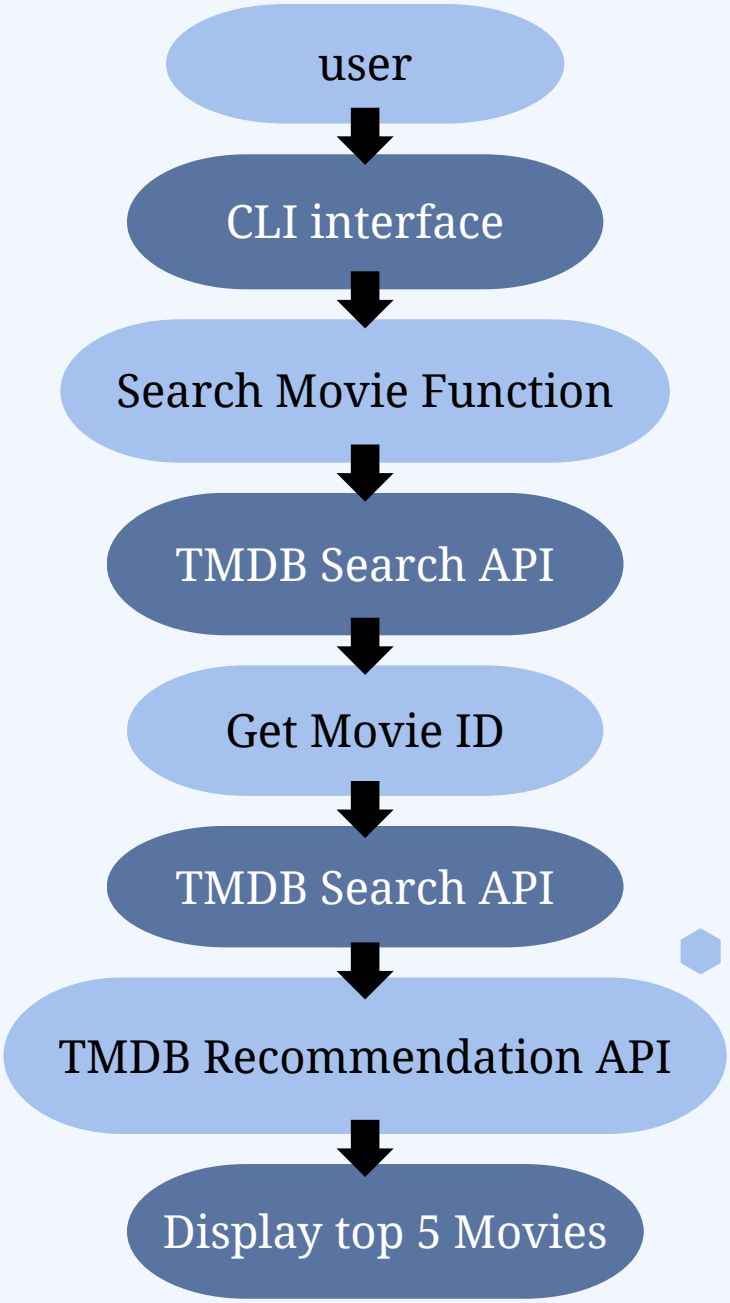
ALGORITHM DEVELOPMENT

Algorithm: API-Based Movie Recommendation

1. Start program.
2. Display menu options.
3. If user selects recommendation:
    - Take movie name as input.
    - Call search_movie(movie_name).
    - If movie ID exists:
        - Call get_recommendations(movie_id).
        - Display top 5 movies.
    - Else:
        - Display “Movie not found”.
4. If user selects exit:
    - Terminate program.
5. Stop.

SOFTWARE ARCHITECTURAL DIAGRAM

Architecture Flow:



CODING

The project is implemented in Python using the requests library.

Key Implementation Points:

- API key authentication.
- HTTP GET requests.
- JSON parsing.
- Exception handling using try-except.
- Loop-based menu system.
- Conditional statements for validation.

OUTPUT

Example Execution:

Movie Recommendation System

1. Get Recommendations
2. Exit

Enter your choice: 1
Enter movie name: Inception

Recommended Movies:

- Interstellar
- The Dark Knight
- Tenet
- Shutter Island
- The Prestige

The system continues running until the user selects Exit.

KEY IMPLEMENTATIONS OUTLINE OF THE SYSTEM

- API Integration with TMDB
- JSON Data Handling
- Error Handling Mechanism
- CLI Menu System
- Real-time Data Fetching

SIGNIFICANT PROJECT OUTCOMES

- Successfully integrated TMDB API.
- Built real-time movie recommendation system.
- Learned HTTP request handling.
- Understood API authentication and response parsing.
- Developed structured Python program.

TESTING AND REFINEMENT

Testing involved:

- Checking valid movie names.
- Testing invalid movie input.
- Testing internet disconnection handling.
- Verifying correct extraction of movie ID.
- Ensuring only top 5 movies are displayed.

Refinements:

- Added exception handling.
- Limited output to 5 recommendations.
- Improved menu structure.

PROJECT APPLICABILITY IN REAL-WORLD APPLICATIONS

- Streaming platforms
- Online entertainment apps
- Movie review websites
- OTT content discovery platforms
- Smart TV recommendation systems

CONTRIBUTION / FINDINGS OF THE PROJECT

- Demonstrated practical API integration.
- Showed how real-world systems use REST APIs.
- Understood dependency on external services.
- Learned structured programming and modular design.

LIMITATION / CONSTRAINTS OF THE SYSTEM

- Requires internet connection.
- Depends on TMDB API availability.
- Limited to top 5 recommendations.
- No graphical interface.
- API rate limit restrictions.

CONCLUSIONS

Here is a more natural and human-friendly version of your conclusion:
The Movie Recommendation System developed using the TMDB API effectively provides real-time movie suggestions based on the user's input. The project demonstrates practical knowledge of API integration, JSON data processing, and modular programming concepts. Through this system, we gain a better understanding of how AI-powered platforms deliver intelligent recommendations using external data services in real-world applications.

FUTURE ENHANCEMENTS

- Convert CLI to Web App using Flask or Streamlit.
- Add movie posters using image URLs.
- Implement user login system.
- Store search history.
- Add genre-based filtering.
- Deploy on cloud platform.

REFERENCES

- TMDB Official API Documentation
- Python requests Library Documentation
- REST API Concepts
- AI & ML Course Materials